

Étale morphisms

SGA 1, Exposé I

English draft

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About this draft

This is an unofficial English translation of Exposé I, “Morphismes étales”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the opening convention and all eleven sections, including their proofs and footnotes. Statement numbering and mathematical notation follow the source, including its two statements numbered 9.2 (a corollary and a proposition). References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé I. Étale morphisms	1
1. Notions of differential calculus	1
2. Quasi-finite morphisms	1
3. Unramified or net morphisms	1
4. Étale morphisms. Étale coverings	3
5. The fundamental property of étale morphisms	4
6. Application to étale extensions of complete local rings	6
7. Local construction of unramified and étale morphisms	6
8. Infinitesimal lifting of étale schemes. Application to formal schemes	8
9. Permanence properties	9
10. Étale coverings of a normal scheme	12
11. Some complements	15

EXPOSÉ I

Étale morphisms

To simplify the exposition, all the preschemes considered are assumed to be locally noetherian, at least after no. 2.

1. Notions of differential calculus

Let X be a prescheme over Y , and let $\Delta_{X/Y}$ or Δ be the diagonal morphism $X \rightarrow X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subprescheme corresponding to the diagonal in V (N.B. if one wants to proceed intrinsically, without assuming X separated over Y —an assumption that would be quite a joke—one should consider the set-theoretic inverse image of $\mathcal{O}_{X \times X}$ on X , and denote by $\mathcal{I}_X$ the augmentation ideal in the latter...). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ can be regarded as a quasi-coherent sheaf on X ; it is denoted by $\Omega_{X/Y}^1$. It is of finite type if $X \rightarrow Y$ is of finite type. It behaves well with respect to extension of the base $Y' \rightarrow Y$. One also introduces the sheaves $\mathcal{O}_{X \times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}_{X/Y}^n$; these are sheaves of *rings* on X , making X into a prescheme which one can denote by $\Delta_{X/Y}^n$ and call the *n*th infinitesimal neighborhood of X/Y . The corresponding sorites is utterly trivial, although rather long¹; it would be prudent to discuss it only when one has something useful to say about it, in connection with smooth morphisms.

2. Quasi-finite morphisms

PROPOSITION 2.1. *Let $A \rightarrow B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A . The following conditions are equivalent:*

- (i) $B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.
- (ii) $\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = k(B)$ is an extension of $k = k(A)$.
- (iii) The completion $\hat{B}$ is finite over the completion $\hat{A}$ of A .

One then says that B is *quasi-finite* over A . A morphism $f: X \rightarrow Y$ is said to be quasi-finite at x (or the Y -prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This also amounts to saying that x is *isolated in its fiber* $f^{-1}(x)$. A morphism is said to be quasi-finite if it is so at every point².

COROLLARY 2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual sorites (i), (ii), (iii), (iv), (v), for quasi-finite morphisms, but this does not seem indispensable here.

3. Unramified or net morphisms

PROPOSITION 3.1. *Let $f: X \rightarrow Y$ be a morphism of finite type, $x \in X$, $y = f(x)$. The following conditions are equivalent:*

- (i) $\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ is a finite separable extension of $k(y)$.
- (ii) $\Omega_{X/Y}^1$ vanishes at x .
- (iii) The diagonal morphism $\Delta_{X/Y}$ is an open immersion in a neighborhood of x .

¹cf. EGA IV 16.3.

²In EGA II 6.2.3, f is also assumed to be of finite type.

For the implication (i) $\Rightarrow$ (ii), Nakayama immediately reduces one to the case where $Y = \text{Spec}(k)$, $X = \text{Spec}(k')$, where this is well known and, moreover, trivial from the definition of separability; (ii) $\Rightarrow$ (iii) follows from a pleasant and easy characterization of open immersions, using Krull; (iii) $\Rightarrow$ (i) because one is again reduced to the case where $Y = \text{Spec}(k)$ and the diagonal morphism is an open immersion everywhere. One must then prove that X is finite with ring separable over k ; for this one is reduced to the case where k is algebraically closed. But then every closed point of X is isolated (being identical to the inverse image of the diagonal under the morphism $X \rightarrow X \times_k X$ defined by x), whence X is finite. One may then suppose that X consists of a single point, with ring A , so $A \otimes_k A \rightarrow A$ is an isomorphism, whence $A = k$, qed.

DEFINITION 3.2. a) One then says that f is *net*, or *unramified*, at x , or that X is net, or unramified, at x over Y .

b) Let $A \rightarrow B$ be a local homomorphism. One says that it is *net*, or *unramified*, or that B is a *net*, or *unramified*, local algebra over A , if $B/\mathfrak{r}(A)B$ is a finite separable extension of $A/\mathfrak{r}(A)$, i.e. if $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $k(B)$ is a separable extension of $k(A)$ ³.

REMARKS. The fact that B is net over A can already be recognized from the completions of A and B . Net implies quasi-finite.

COROLLARY 3.3. *The set of points where f is net is open.*

COROLLARY 3.4. *Let X', X be two preschemes of finite type over Y , and $g: X' \rightarrow X$ a Y -morphism. If X is net over Y , the graph morphism $\Gamma_g: X' \rightarrow X \times_Y X$ is an open immersion.*

Indeed, it is the inverse image of the diagonal morphism $X \rightarrow X \times_Y X$ under

$$g \times_Y \text{id}_{X'}: X' \times_Y X \longrightarrow X \times_Y X.$$

One can also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega_{X/Y}^1$, called the different ideal of X/Y ; it defines a closed subprescheme of X whose underlying set is the set of points where X/Y is ramified, i.e. not net.

PROPOSITION 3.5. (i) *An immersion is net.*

(ii) *The composite of two net morphisms is net.*

(iii) *Extension of the base in a net morphism gives another net morphism.*

This can be seen equally well from (ii) or (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of definition b)), and tedious. As usual, one obtains corollaries:

COROLLARIES 3.6. (iv) *A cartesian product of two net morphisms is another net morphism.*

(v) *If gf is net, f is net.*

(vi) *If f is net, f_{red} is net.*

PROPOSITION 3.7. *Let $A \rightarrow B$ be a local homomorphism; suppose that the residue field extension $k(B)/k(A)$ is trivial or that $k(A)$ is algebraically closed. For B/A to be net, it is necessary and sufficient that $\hat{B}$ be (as an $\hat{A}$ -algebra) a quotient of $\hat{A}$.*

REMARKS. – When the residue field extension is not assumed trivial, one can reduce to this case by making a suitable finite flat extension of A which kills the said extension.

– Give the example where A is the local ring of an ordinary double point of a curve, and B that of a point of its normalization: then $A \subset B$, B is net over A with trivial residue field extension, and $\hat{A} \rightarrow \hat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of being net.

³Cf. the second thoughts in III 1.2.

4. Étale morphisms. Étale coverings

We shall admit everything we shall need concerning flat morphisms; these facts will be proved later, if necessary⁴.

- DEFINITION 4.1. a) Let $f: X \rightarrow Y$ be a morphism of finite type. One says that f is *étale* at x if f is flat at x and net at x . One says that f is *étale* if it is so at every point. One says that X is *étale* at x over Y , or that it is a Y -prescheme *étale* at x , etc.
- b) Let $f: A \rightarrow B$ be a local homomorphism. One says that f is *étale*, or that B is *étale* over A , if B is flat and unramified over A .⁵

PROPOSITION 4.2. *For B/A to be étale, it is necessary and sufficient that $\hat{B}/\hat{A}$ be so.*

In fact, this is true separately for “net” and for “flat”.

COROLLARY 4.3. *Let $f: X \rightarrow Y$ be of finite type, and $x \in X$. Whether f is étale at x depends only on the local homomorphism $\mathcal{O}_{f(x)} \rightarrow \mathcal{O}_x$, and indeed only on the corresponding homomorphism of completions.*

COROLLARY 4.4. *Suppose that the residue field extension $k(A) \rightarrow k(B)$ is trivial, or that $k(A)$ is algebraically closed. Then B/A is étale if and only if $\hat{A} \rightarrow \hat{B}$ is an isomorphism.*

Combine flatness and 3.7.

PROPOSITION 4.5. *Let $f: X \rightarrow Y$ be a morphism of finite type. Then the set of points where it is étale is open.*

Indeed, this is true separately for “net” and “flat”.

This proposition shows that one can dispense with “pointwise” statements in studying morphisms of finite type which are étale somewhere.

- PROPOSITION 4.6. (i) *An open immersion is étale.*
(ii) *The composite of two étale morphisms is étale.*
(iii) *Extension of the base.*

Indeed, (i) is trivial, and for (ii) and (iii) it suffices to note that this is true for “net” and for “flat”. To tell the truth, there are also corresponding statements for local homomorphisms (without a finiteness condition), which in any case will have to appear in the multiplodogue (beginning with the case: net).

COROLLARY 4.7. *A cartesian product of two étale morphisms is likewise étale.*

COROLLARY 4.8. *Let X and X' be of finite type over Y , and $g: X \rightarrow X'$ a Y -morphism. If X' is unramified over Y and X is étale over Y , then g is étale.*

Indeed, g is the composite of the graph morphism $\Gamma_g: X \rightarrow X \times_Y X'$, which is an open immersion by 3.4, and the projection morphism, which is étale because it is obtained from the étale morphism $X \rightarrow Y$ by the “change of base” $X' \rightarrow Y$.

DEFINITION 4.9. An étale (resp. net) covering of Y is a Y -scheme X which is finite over Y and étale (resp. net) over Y .

The first condition means that X is defined by a coherent sheaf of algebras $\mathcal{B}$ on Y . The second then means that $\mathcal{B}$ is locally free on Y (resp. nothing at all), and that, in addition, for every $y \in Y$, the fiber $\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_y} k(y)$ is a separable algebra (= finite product of finite separable extensions) over $k(y)$.

⁴Cf. Exp. IV.

⁵Cf. the second thoughts in III 1.2.

PROPOSITION 4.10. *Let X be a flat covering of Y of degree n (the definition of this term deserved to appear in 4.9), defined by a coherent locally free sheaf $\mathcal{B}$ of algebras. One defines in the well-known way the trace homomorphism $\mathcal{B} \rightarrow \mathcal{A}$ (which is a homomorphism of $\mathcal{A}$ -modules, where $\mathcal{A} = \mathcal{O}_Y$). For X to be étale, it is necessary and sufficient that the corresponding bilinear form $\mathrm{tr}_{\mathcal{B}/\mathcal{A}} xy$ define an isomorphism of $\mathcal{B}$ onto $\mathcal{B}$, or, what amounts to the same, that the discriminant section*

$$d_{X/Y} = d_{\mathcal{B}/\mathcal{A}} \in \Gamma(Y, \bigwedge^n \check{\mathcal{B}} \otimes_{\mathcal{A}} \bigwedge^n \check{\mathcal{B}})$$

be invertible, or finally that the discriminant ideal defined by this section be the unit ideal.

Indeed, one is reduced to the case where $Y = \mathrm{Spec}(k)$, and then this is a well-known separability criterion, trivial on passing to the algebraic closure of k .

REMARK. We shall have a less trivial statement below, when one does not assume a priori that X is flat over Y but makes a normality assumption.

5. The fundamental property of étale morphisms

THEOREM 5.1. *Let $f: X \rightarrow Y$ be a morphism of finite type. For f to be an open immersion, it is necessary and sufficient that it be an étale and radicial morphism.*

Recall that radicial means: injective, with purely inseparable residue field extensions (and one may also recall that this means that the morphism remains injective under every extension of the base). Necessity is trivial; sufficiency remains. We shall give two different proofs, the first shorter, the second more elementary.

1) A flat morphism is open, so one may suppose (replacing Y by $f(X)$) that f is a homeomorphism onto Y . Under any extension of the base, f will remain flat, radicial, and surjective, hence a homeomorphism, a fortiori closed. Thus f is *proper*. Thus f is *finite* (reference: Chevalley's theorem), defined by a coherent sheaf $\mathcal{B}$ of algebras. $\mathcal{B}$ is locally free; moreover, by the hypothesis it has rank 1 everywhere, so $X = Y$, qed.

2) One may suppose that Y and X are *affine*. One is further easily reduced to proving the following: if $Y = \mathrm{Spec}(A)$, with A local, and if $f^{-1}(y)$ is nonempty (y being the closed point of Y), then $X = Y$ (indeed, this will imply that every $y \in f(X)$ has an open neighborhood U such that $X|_U = U$). We shall have $X = \mathrm{Spec}(B)$; we want to prove that $A = B$. But for this one is reduced to proving the analogous assertion on replacing A by $\hat{A}$, and B by $B \otimes_A \hat{A}$ (taking into account that $\hat{A}$ is faithfully flat over A). One may therefore suppose that A is *complete*. Let x be the point above y ; by Corollary 2.2, $\mathcal{O}_x$ is finite over A and hence (being flat and radicial over A) is identical to A . Thus $X = Y \amalg X'$ (disjoint sum). Since X is radicial over Y , X' is empty. We are done.

COROLLARY 5.2. *Let $f: X \rightarrow Y$ be a morphism which is a closed immersion and is étale. If X is connected, f is an isomorphism of X onto a connected component of Y .*

Indeed, f is also an open immersion. One deduces:

COROLLARY 5.3. *Let X be a net Y -scheme, with Y connected. Then every section of X over Y is an isomorphism of Y onto a connected component of X . There is therefore a one-to-one correspondence between the set of these sections and the set of connected components X_i of X such that the projection $X_i \rightarrow Y$ is an isomorphism (or, equivalently by 5.1, is surjective and radicial). In particular, a section is known when its value at one point is known.*

Only the first assertion requires a proof; by 5.2, it suffices to observe that a section is a closed immersion (since X is separated over Y) and is étale by 4.8.

COROLLARY 5.4. *Let X and Y be two preschemes over S , with X net and separated over S and Y connected. Let f, g be two S -morphisms from Y to X , and y a point of Y ; suppose that $f(y) = g(y) = x$ and that the residue field homomorphisms $k(x) \rightarrow k(y)$ defined by f and g are identical ("f and g agree geometrically at y"). Then f and g are identical.*

This follows from 5.3 by reducing to the case where $Y = S$, replacing X by $X \times_S Y$. Here is a particularly important variant of 5.3.

THEOREM 5.5. *Let S be a prescheme, X and Y two S -preschemes, S_0 a closed subscheme of S with the same underlying space as S , and $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$ the “restrictions” of X and Y to S_0 . Suppose that X is étale over S . Then the natural map*

$$\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_{S_0}(Y_0, X_0)$$

is bijective.

One is again reduced to the case where $Y = S$, and then this follows from the “topological” description of the sections of X/Y given in 5.3.

SCHOLIUM. This result contains an assertion of *uniqueness* and *existence of morphisms*. It can also be expressed (when X and Y are both taken étale over S) by saying that the functor $X \mapsto X_0$ from the category of étale S -schemes to the category of étale S_0 -schemes is *fully faithful*, i.e. establishes an *equivalence* of the first with a *full subcategory* of the second. We shall see below that it is even an equivalence of the first and the second (which will be a theorem on the *existence of étale S -schemes*).

The following form of 5.5, more general in appearance, is often convenient:

COROLLARY 5.6 (“Theorem on extending liftings”). *Consider a commutative diagram*

$$\begin{array}{ccc} X & \longleftarrow & Y_0 \\ \downarrow & & \downarrow \\ S & \longleftarrow & Y \end{array}$$

of morphisms, where $X \rightarrow S$ is étale and $Y_0 \rightarrow Y$ is a bijective closed immersion. Then one can find a unique morphism $Y \rightarrow X$ which makes the two corresponding triangles commutative.

Indeed, replacing S by Y and X by $X \times_S Y$, one is reduced to the case where $Y = S$, and then this is the special case of 5.5 for $Y = S$.

Let us also point out the following immediate consequence of 5.1 (which we did not give as Corollary 1 so as not to interrupt the line of ideas developed after 5.1):

PROPOSITION 5.7. *Let X, X' be two preschemes of finite type and flat over Y , and let $g: X \rightarrow X'$ be a Y -morphism. For g to be an open immersion (resp. an isomorphism), it is necessary and sufficient that, for every $y \in Y$, the induced morphism on the fibers*

$$g \otimes_Y k(y): X \otimes_Y k(y) \longrightarrow X' \otimes_Y k(y)$$

be so.

It suffices to prove sufficiency; since the assertion holds for surjectivity, one is reduced to the case of an open immersion. By 5.1, one must check that g is *radicial*, which is trivial, and that it is *étale*, which follows from Corollary 5.9 below.

COROLLARY 5.8. *(Should be moved to no. 3.) Let X and X' be two Y -preschemes, $g: X \rightarrow X'$ a Y -morphism, x a point of X , and y its projection on Y . For g to be quasi-finite (resp. net) at x , it is necessary and sufficient that $g \otimes_Y k(y)$ be so.*

Indeed, the two algebras over $k(g(x))$ which one must examine to check that one indeed has a morphism quasi-finite resp. net at x are the same for g and $g \otimes_Y k(y)$.

COROLLARY 5.9. *With the notation of 5.8, suppose that X and X' are flat and of finite type over Y . For g to be flat (resp. étale) at x , it is necessary and sufficient that $g \otimes_Y k(y)$ be so.*

For “flat”, the statement is included only as a reminder; it is one of the fundamental flatness criteria⁶. For étale, it follows from this, taking 5.8 into account.

⁶Cf. IV 5.9.

6. Application to étale extensions of complete local rings

This no. is a special case of results on formal preschemes which will have to appear in the multiplodoque. Nevertheless, one gets by here with less effort, i.e. without the explicit local determination of étale morphisms in no. 7 (using the Main Theorem). This is perhaps sufficient reason to keep the present no. (even in the multiplodoque) in this place.

THEOREM 6.1. *Let A be a complete local ring (noetherian, of course), with residue field k . For every A -algebra B , let $R(B) = B \otimes_A k$, regarded as a k -algebra; it therefore depends functorially on B . Then R defines an equivalence of the category of A -algebras finite and étale over A with the category of separable algebras of finite rank over k .*

First of all, the functor in question is fully faithful, as follows from the more general fact:

COROLLARY 6.2. *Let B, B' be two A -algebras finite over A . If B is étale over A , then the canonical map*

$$\mathrm{Hom}_{A\text{-alg}}(B, B') \longrightarrow \mathrm{Hom}_{k\text{-alg}}(R(B), R(B'))$$

is bijective.

One is reduced to the case where A is artinian (by replacing A by $A/\mathfrak{m}^n$), and then this is a special case of 5.5.

It remains to prove that for every finite separable k -algebra (why not say: étale, it is shorter) L , there exists a B étale over A such that $R(B)$ is isomorphic to L . One may suppose that L is a separable extension of k ; as such, it has a generator x , i.e. is isomorphic to an algebra $k[t]/Fk[t]$ where $F \in k[t]$ is a monic polynomial. One lifts F to a monic polynomial F_1 in $A[t]$, and takes $B = A[t]/F_1A[t]$.

7. Local construction of unramified and étale morphisms

PROPOSITION 7.1. *Let A be a noetherian ring, B a finite algebra over A , u a generator of B over A , $F \in A[t]$ such that $F(u) = 0$ (F is not assumed monic), $u' = F'(u)$ (where F' is the derivative polynomial), $\mathfrak{q}$ a prime ideal of B not containing u' , and $\mathfrak{p}$ its contraction to A . Then $B_{\mathfrak{q}}$ is net over $A_{\mathfrak{p}}$.*

In other words, setting $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $X_{u'} = \mathrm{Spec}(B_{u'})$, $X_{u'}$ is unramified over Y . The statement follows from the following, more precise one:

COROLLARY 7.2. *The different ideal of B/A contains $u'B$, and is equal to it if the natural homomorphism $A[t]/FA[t] \rightarrow B$ (sending t to u) is an isomorphism.*

Let J be the kernel of the homomorphism $C = A[t] \rightarrow B$; this kernel contains $FA[t]$, and is equal to it in the second case considered in 7.2. Since the homomorphism is surjective, $\Omega_{B/A}^1$ is identified with the quotient of $\Omega_{C/A}^1$ by the submodule generated by $J\Omega_{C/A}^1$ and $d(J)$ (the definition of the homomorphism d , and the calculation of Ω^1 for a polynomial algebra, should have been made explicit in no. 1). Identifying $\Omega_{C/A}^1$ with C by means of the basis dt , one finds $B/B \cdot J'$; thus the different is generated by the set J' of images in B of the derivatives of the $G \in J$ (and it suffices to take G generating J). Since $F \in J$, resp. F is a generator of J , we are done. (N.B. One should make 7.2 a proposition and 7.1 a corollary.) One obtains:

COROLLARY 7.3. *Under the conditions of 7.1, assuming that F is monic and that $A[t]/FA[t] \rightarrow B$ is an isomorphism, for $B_{\mathfrak{q}}$ to be étale over $A_{\mathfrak{p}}$, it is necessary and sufficient that $\mathfrak{q}$ not contain u' .*

Indeed, since B is flat over A , étale is equivalent to net, and one can apply 7.2.

COROLLARY 7.4. *Under the conditions of 7.3, for B to be étale over A , it is necessary and sufficient that u' be invertible, or equivalently that the ideal generated by F, F' in $A[t]$ be the unit ideal.*

The last criterion follows from the first and Nakayama (in B).

A monic polynomial $F \in A[t]$ having the property stated in Corollary 7.4 is called a *separable polynomial* (if F is not monic, one should at least require its leading coefficient to be invertible; when A is a field, one recovers the usual definition).

COROLLARY 7.5. *Let B be a finite algebra over the local ring A . Suppose that $K(A)$ is infinite or that B is local. Let n be the rank of $L = B \otimes_A K(A)$ over $K(A) = k$. For B to be net (resp. étale) over A , it is necessary and sufficient that B be isomorphic to a quotient of (resp. isomorphic to) $A[t]/FA[t]$, where F is a monic separable polynomial which one may assume to be (resp. which is necessarily) of degree n .*

Only necessity needs to be proved. Suppose that B is net over A , so that L is separable over k . It then follows from the hypothesis that L/k has a generator ξ , so the ξ^i ($0 \leq i < n$) form a basis of L over k . Let $u \in B$ lift ξ ; then, by Nakayama, the u^i ($0 \leq i < n$) generate the A -module B (resp. form a basis of it). In particular, one can find a monic polynomial $F \in A[t]$ such that $F(u) = 0$, and B will be isomorphic to a quotient of (resp. isomorphic to) $A[t]/FA[t]$. Finally, by 7.4 applied to L/k , F and F' generate $A[t]$ modulo $\mathfrak{m}A[t]$; hence (by Nakayama in $A[t]/FA[t]$) F and F' generate $A[t]$, and we are done.

THEOREM 7.6. *Let A be a local ring, and $A \rightarrow \mathcal{O}$ a local homomorphism such that $\mathcal{O}$ is isomorphic to a localization of an algebra of finite type over A . Suppose that $\mathcal{O}$ is net over A . Then one can find an A -algebra B , integral over A , a maximal ideal $\mathfrak{n}$ of B , a generator u of B over A , and a monic polynomial $F \in A[t]$, such that $\mathfrak{n} \not\supset F'(u)$ and $\mathcal{O}$ is isomorphic (as an A -algebra) to $B_{\mathfrak{n}}$. If $\mathcal{O}$ is étale over A , one can take $B = A[t]/FA[t]$.*

(Of course, these conditions are also sufficient...)

Let us first point out the pleasant corollaries:

COROLLARY 7.7. *For $\mathcal{O}$ to be net over A , it is necessary and sufficient that $\mathcal{O}$ be isomorphic to the quotient of an analogous algebra which is étale over A .*

Indeed, one takes $\mathcal{O}' = B'_{\mathfrak{n}'}$, where $B' = A[t]/FA[t]$ and $\mathfrak{n}'$ is the inverse image of $\mathfrak{n}$ in B' .

COROLLARY 7.8. *Let $f: X \rightarrow Y$ be a morphism of finite type, $x \in X$. For f to be net at x , it is necessary and sufficient that there exist an open neighborhood U of x such that $f|_U$ factors as $U \rightarrow X' \rightarrow Y$, where the first arrow is a closed immersion and the second an étale morphism.*

This is a mere translation of 7.7.

Let us show how the jargon of 7.6 follows from the main statement: indeed, by 7.7 there is an epimorphism $\mathcal{O}' \rightarrow \mathcal{O}$, where $\mathcal{O}$ has the desired properties; but since $\mathcal{O}'$ and $\mathcal{O}$ are étale over A , the morphism $\mathcal{O}' \rightarrow \mathcal{O}$ is étale by 4.8, hence an isomorphism.

Proof of 7.6. It follows a proof from the Chevalley seminar. By the *Main Theorem*, one has $\mathcal{O} = B_{\mathfrak{n}}$, where B is a finite algebra over A and $\mathfrak{n}$ is a maximal ideal of it. Then $B/\mathfrak{n} = K(\mathcal{O})$ is a separable, hence monogenic, extension of k ; if $\mathfrak{n}_i$ ($1 \leq i \leq r$) are the maximal ideals of B distinct from $\mathfrak{n}$, there is therefore an element u of B which belongs to all the $\mathfrak{n}_i$, and whose image in $B/\mathfrak{n}$ is a generator of it. Now $B/\mathfrak{n} = B_{\mathfrak{n}}/\mathfrak{n}B_{\mathfrak{n}} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ (where $\mathfrak{m}$ is the maximal ideal of A). Let us admit for a moment the following

LEMMA 7.9. *Let A be a local ring, B a finite algebra over A , $\mathfrak{n}$ a maximal ideal of B , and u an element of B whose image in $B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ generates it as an algebra over $k = A/\mathfrak{m}$, and which belongs to every maximal ideal of B distinct from $\mathfrak{n}$. Let $B' = B[u]$, $\mathfrak{n}' = \mathfrak{n}B'$. Then the canonical homomorphism $B'_{\mathfrak{n}'} \rightarrow B_{\mathfrak{n}}$ is an isomorphism.*

LEMMA 7.10. *(Should have appeared as a corollary to 7.1, before 7.5, which it implies.) Let B be a finite algebra over A generated by an element u , and let $\mathfrak{n}$ be a maximal ideal of B such that $B_{\mathfrak{n}}$ is unramified over A . Then there exists a monic polynomial $F \in A[t]$ such that $F(u) = 0$ and $F'(u) \notin \mathfrak{n}$.*

Indeed, let n be the rank of the k -algebra $L = B \otimes_A k$; by Nakayama there exists a monic polynomial of degree n in $A[t]$ such that $F(u) = 0$. Let f be the polynomial obtained from F by reduction mod $\mathfrak{m}$; then L is k -isomorphic to $k[t]/fk[t]$, so by 7.3, $f'(\xi)$ does not belong to the maximal ideal of L corresponding to $\mathfrak{n}$ (ξ denoting the image of t in L , i.e. the image of u in L). Since $f'(\xi)$ is the image of $F'(u)$, we are done.

The theorem 7.6 now follows by combining 7.9 and 7.10. It remains to prove 7.9. Set $S' = B' - \mathfrak{n}'$, so $B'S'^{-1} = B'_{\mathfrak{n}'}$.

$$\begin{array}{ccccc} B & \longrightarrow & BS'^{-1} & \longrightarrow & BS^{-1} = B_{\mathfrak{n}} \\ \uparrow & & \uparrow & & \\ B' & \longrightarrow & B'S'^{-1} = B'_{\mathfrak{n}'} & & \end{array}$$

Similarly, let $S = B - \mathfrak{n}$, so $BS^{-1} = B_{\mathfrak{n}}$. There is therefore a natural homomorphism $BS'^{-1} \rightarrow BS^{-1} = B_{\mathfrak{n}}$; let us prove that it is an isomorphism, i.e. that the elements of S are invertible in BS'^{-1} , i.e. that every maximal ideal $\mathfrak{p}$ of the latter is disjoint from S , i.e. contracts to $\mathfrak{n}$ in B . Indeed, since BS'^{-1} is finite over $B'S'^{-1} = B'_{\mathfrak{n}'}$, $\mathfrak{p}$ contracts to the unique maximal ideal $\mathfrak{n}'B'_{\mathfrak{n}'}$ of $B'_{\mathfrak{n}'}$, hence contracts to the maximal ideal $\mathfrak{n}'$ of B' ; since B is finite over B' , the ideal $\mathfrak{q}$ of B induced by $\mathfrak{p}$, lying over $\mathfrak{n}'$, is necessarily maximal, and does not contain u , hence is identical to $\mathfrak{n}$. (We have just used the fact that u belongs to every maximal ideal of B distinct from $\mathfrak{n}$.) Let us now prove that BS'^{-1} equals $B'S'^{-1}$: since it is finite over the latter, Nakayama reduces one to proving equality mod $\mathfrak{n}'BS'^{-1}$, and a fortiori it suffices to prove equality mod $\mathfrak{m}BS'^{-1}$; now $BS'^{-1}/\mathfrak{m}BS'^{-1} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ is generated over k by u (here one uses the other property of u), so the image of B' (and a fortiori of $B'S'^{-1}$) in it is the whole ring (being a subring containing k and the image of u).

REMARK. It should be possible to state Theorem 7.6 for a ring $\mathcal{O}$ which is merely semilocal, so as to include 7.5 as well: one would assume that $\mathcal{O}/\mathfrak{m}\mathcal{O}$ is a *monogenic* k -algebra; one could therefore find a $u \in B$ whose image in $B/\mathfrak{m}B$ is a generator, and which belongs to all the maximal ideals of B not coming from $\mathcal{O}$. Lemmas 7.9 and 7.10 should adapt without difficulty. More generally, ...

8. Infinitesimal lifting of étale schemes. Application to formal schemes

PROPOSITION 8.1. *Let Y be a prescheme, Y_0 a subprescheme, X_0 an étale Y_0 -scheme, and x a point of X_0 . Then there exist an étale Y -scheme X , a neighborhood U_0 of x in X_0 , and a Y_0 -isomorphism $U_0 \xrightarrow{\sim} X \times_Y Y_0$.*

Indeed, let y be the projection of x on Y_0 ; applying 7.6 to the étale local homomorphism $A_0 \rightarrow B_0$ of the local rings of y and x in Y_0 and X_0 , one finds an isomorphism

$$B_0 = C_{0\mathfrak{n}_0} \quad C_0 = A_0[t]/F_0A_0[t]$$

where F_0 is a monic polynomial and $\mathfrak{n}_0$ is a maximal ideal of C_0 not containing the class of $F'_0(t)$ in C_0 . Let A be the local ring of y in Y , let F be a monic polynomial in $A[t]$ giving F_0 under the surjective homomorphism $A \rightarrow A_0$ (one lifts the coefficients of F_0), and finally let $C = A[t]/FA[t]$ and $\mathfrak{n}$ be the maximal ideal of C which is the inverse image of $\mathfrak{n}_0$ under the natural epimorphism $C \rightarrow C \otimes_A A_0 = C_0$. Set

$$B = C_{\mathfrak{n}}.$$

It is immediate from the construction and 7.1 that B is étale over A , and that there is an isomorphism $B \otimes_A A_0 = A_0$. One knows (Chap. I) that there exist a Y -scheme X of finite type and a point z of X above y such that $\mathcal{O}_z$ is A -isomorphic to C ; since the latter is étale over $A = \mathcal{O}_y$, one may (taking X small enough) suppose that X is étale over Y . Let $X'_0 = X \times_Y Y_0$; then the local ring of z in X'_0 is identified with $\mathcal{O}_z \otimes_A A_0 = B \otimes_A A_0$, hence is isomorphic to B_0 . This isomorphism is defined by an isomorphism of a neighborhood U_0

of x in X_0 onto a neighborhood of z in X'_0 (loc. cit.), which one may suppose identical to X'_0 by taking X small enough. We are done.

COROLLARY 8.2. *An analogous statement holds for étale coverings, assuming that the residue field $k(y)$ is infinite.*

The proof is the same, with 7.5 replacing 7.6.

THEOREM 8.3. *The functor considered in 5.5 is an equivalence of categories.*

By Theorem 5.5, it remains to show that every étale S_0 -scheme X_0 is isomorphic to an S_0 -scheme $X \times_S S_0$, where X is an étale S -scheme. The topological space underlying X must necessarily be identical to that of X_0 , with X_0 moreover identified with a closed subscheme of X . The problem is therefore equivalent to the following: to find, on the topological space $|X_0|$ underlying X_0 , a sheaf of algebras $\mathcal{O}_X$ over $f_0^*(\mathcal{O}_S)$ (where f_0 is the projection $X_0 \rightarrow S_0$, regarded here as a continuous map of the underlying spaces), making $|X_0|$ into an étale S -prescheme X , and a homomorphism of algebras $\mathcal{O}_X \rightarrow \mathcal{O}_{X_0}$, compatible with the homomorphism $f_0^*(\mathcal{O}_S) \rightarrow f_0^*(\mathcal{O}_{S_0})$ of sheaves of scalars, inducing an isomorphism $\mathcal{O}_X \otimes_{f_0^*(\mathcal{O}_S)} f_0^*(\mathcal{O}_{S_0}) \xrightarrow{\sim} \mathcal{O}_{X_0}$. (Then X will be an étale S -prescheme reducing to X_0 , hence will be separated over S since X_0 is separated over S_0 , and X answers the question.) Moreover, if (U_i) is a covering of X_0 by open subsets, and if one has found a solution of the problem in each U_i , it follows from the uniqueness theorem 5.5 that these solutions glue (i.e. the sheaves of algebras defining them, equipped with their augmentation homomorphisms, glue), and one sees at once that the ringed space thus constructed over S is an étale S -prescheme X equipped with an isomorphism $X \times_S S_0 \xrightarrow{\sim} X_0$. It therefore suffices to find a solution locally, which is ensured by 8.1.

COROLLARY 8.4. *Let S be a locally noetherian formal prescheme, equipped with an ideal of definition J , and let $S_0 = (|S|, \mathcal{O}_S/J)$ be the corresponding ordinary prescheme. Then the functor $\mathfrak{X} \mapsto \mathfrak{X} \times_S S_0$ from the category of étale coverings of S to the category of étale coverings of S_0 is an equivalence of categories.*

Of course, by an étale covering of a *formal* prescheme S one means a covering of S , i.e. a formal prescheme over S defined by means of a coherent sheaf of algebras $\mathcal{B}$, such that $\mathcal{B}$ is *locally free* and the residue fibers $\mathcal{B}_s \otimes_{\mathcal{O}_s} k(s)$ of $\mathcal{B}$ are *separable* algebras over $k(s)$. If one denotes by S_n the ordinary prescheme $(|S|, \mathcal{O}_S/J^{n+1})$, the datum of a coherent sheaf of algebras $\mathcal{B}$ on S is equivalent to the datum of a sequence of coherent sheaves of algebras $\mathcal{B}_n$ on the S_n , equipped with a transitive system of homomorphisms $\mathcal{B}_m \rightarrow \mathcal{B}_n$ ($m \geq n$) defining isomorphisms $\mathcal{B}_m \otimes_{\mathcal{O}_{S_m}} \mathcal{O}_{S_n} \xrightarrow{\sim} \mathcal{B}_n$. It is immediate that $\mathcal{B}$ is locally free if and only if the $\mathcal{B}_n$ on the S_n are so, and that the separability condition holds if and only if it holds for $\mathcal{B}_0$, or equivalently for all the $\mathcal{B}_n$. Thus $\mathcal{B}$ is étale over S if and only if the $\mathcal{B}_n$ on the S_n are so. In view of this, 8.4 follows at once from 8.3.

REMARK. It was not necessary in 8.4 to restrict oneself to *coverings*. This is, however, the only case used for the moment.

9. Permanence properties

Let $A \rightarrow B$ be a local étale homomorphism. We examine here some cases in which a certain property of A implies the same property for B , or conversely. A number of such propositions already follow merely from the fact that B is *quasi-finite* and *flat* over A , and we shall confine ourselves to “recalling” a few of them: *A and B have the same Krull dimension and the same depth* (Serre’s “cohomological codimension”, in the terminology still current). It follows, for example, that *A is Cohen–Macaulay if and only if B is so*. Moreover, for every prime ideal $\mathfrak{q}$ of B , inducing $\mathfrak{p}$ on A , $B_{\mathfrak{q}}$ will again be quasi-finite and flat over $A_{\mathfrak{p}}$, provided that B is assumed to be a localization of an algebra of finite type over A (this follows from the fact that the set of points where a morphism of finite type is quasi-finite resp. flat is

open); furthermore, *every* prime ideal $\mathfrak{p}$ of A is induced by a prime ideal $\mathfrak{q}$ of B (since B is faithfully flat over A). It follows, for example, that $\mathfrak{p}$ and $\mathfrak{q}$ have the same height; and also that A has no embedded prime ideals if and only if B has none.

We shall therefore restrict ourselves to propositions more specific to the case of étale morphisms.

PROPOSITION 9.1. *Let $A \rightarrow B$ be a local étale homomorphism. For A to be regular, it is necessary and sufficient that B be so.*

Indeed, let k be the residue field of A , and L that of B . Since B is flat over A and $L = B \otimes_A k$, i.e. $\mathfrak{n} = \mathfrak{m}B$ (where $\mathfrak{m}, \mathfrak{n}$ are the maximal ideals of A, B), the $\mathfrak{m}$ -adic filtration on B is identical to its $\mathfrak{n}$ -adic filtration, and one has

$$\mathrm{gr}^*(B) = \mathrm{gr}^*(A) \otimes_k L.$$

It follows that $\mathrm{gr}^*(B)$ is a polynomial algebra over L if and only if $\mathrm{gr}^*(A)$ is a polynomial algebra over k , qed. (N.B. The fact that L/k is separable has not been used.)

COROLLARY 9.2. *Let $f: X \rightarrow Y$ be an étale morphism. If Y is regular, so is X ; the converse holds if f is surjective.*

PROPOSITION 9.2. *Let $f: X \rightarrow Y$ be an étale morphism. If Y is reduced, so is X ; the converse holds if f is surjective.*

This is equivalent to the following

COROLLARY 9.3. *Let $f: A \rightarrow B$ be a local étale homomorphism, with B isomorphic to an A -algebra which is a localization of an A -algebra of finite type. For A to be reduced, it is necessary and sufficient that B be so.*

Necessity is trivial, since $A \rightarrow B$ is injective (B being faithfully flat over A). Sufficiency: let $\mathfrak{p}_i$ be the minimal prime ideals of A ; by hypothesis the natural map $A \rightarrow \prod_i A/\mathfrak{p}_i$ is injective, so tensoring with the flat A -module B , one finds that $B \rightarrow \prod_i B/\mathfrak{p}_i B$ is injective, and one is reduced to proving that the $B/\mathfrak{p}_i B$ are reduced. Since $B/\mathfrak{p}_i B$ is étale over $A/\mathfrak{p}_i$, one is reduced to the case where A is an integral domain. Let K be its field of fractions; since $A \rightarrow K$ is injective, so is $B \rightarrow B \otimes_A K$ (B being A -flat), and one is reduced to proving that the latter ring is reduced. Now, B , being a localization of an A -algebra of finite type over A , is the local ring of a point x of a scheme $X = \mathrm{Spec}(C)$ of finite type and étale over $Y = \mathrm{Spec}(A)$; hence $B \otimes_A K$ is a localization (with respect to a suitable multiplicatively closed set) of the ring $C \otimes_A K$ of $X \otimes_A K$. Since $X \otimes_A K$ is étale over K , its ring is a finite product of fields (separable extensions of K); hence the same holds for $B \otimes_A K$, qed.

COROLLARY 9.4. *Let $f: A \rightarrow B$ be a local étale homomorphism, and suppose that A is analytically reduced (i.e. the completion $\hat{A}$ of A has no nilpotent elements). Then B is analytically reduced, and a fortiori reduced.*

Indeed, $\hat{B}$ is finite and étale over $\hat{A}$, and one applies 9.3.

THEOREM 9.5. *Let $f: A \rightarrow B$ be a local homomorphism, with B isomorphic to a localization of an A -algebra of finite type. Then:*

- (i) *If f is étale, A is normal if and only if B is so.*
- (ii) *If A is normal, f is étale if and only if f is injective and net (and then B is normal by (i)).*

We shall give two different proofs of (i). The first uses some of the properties of flat quasi-finite morphisms (recalled at the beginning of this no.) without using 7.6 (and thereby the Main Theorem); the reverse is true of the second proof. Finally, for (ii) it seems that one needs the Main Theorem in any case.

First proof. One uses the following necessary and sufficient condition for normality of a noetherian local ring A of dimension $\neq 0$.

SERRE'S CRITERION. (i) *For every prime ideal $\mathfrak{p}$ of A of height 1, $A_{\mathfrak{p}}$ is normal (or, equivalently, regular);* (ii) *For every prime ideal $\mathfrak{p}$ of A of height ≥ 2 , the depth of $A_{\mathfrak{p}}$ is ≥ 2 .*⁷

We shall admit this criterion here; it is supposed to appear in the paragraph on flat morphisms. Its main advantage is that it does not assume a priori that A is reduced, let alone an integral domain. Here one may already suppose that $\dim A = \dim B \neq 0$.

By the facts recalled at the beginning of this no., the prime ideals $\mathfrak{p}$ of A of height 1 (resp. of height ≥ 2) are precisely the contractions to A of the prime ideals $\mathfrak{q}$ of B of height 1 (resp. of height ≥ 2). Finally, if $\mathfrak{p}$ and $\mathfrak{q}$ correspond, $B_{\mathfrak{q}}$ is étale over $A_{\mathfrak{p}}$, hence has the same depth as $A_{\mathfrak{p}}$, and is regular if and only if $A_{\mathfrak{p}}$ is so (9.1). Applying Serre's criterion, one finds that A is normal if and only if B is so.

Second proof. Suppose that B is normal; let L be its field of fractions, and K that of A (A is an integral domain since B is so). We saw in the proof of 9.3 that $B \otimes_A K$ is a finite product of fields; since it is contained in L it is a field, and since it contains B it is L . An element of K integral over A is integral over B , hence belongs to B since B is normal, and therefore to A since $B \cap K = A$ (as follows from the fact that B is faithfully flat over A).

Suppose now that A is normal; let us prove that B is so. By 7.6, one has $B = B'_{\mathfrak{n}}$, where $B' = A[t]/FA[t]$, with F and $\mathfrak{n}$ as in 7.6. Thus $L = B \otimes_A K$ will be a localization of $B' \otimes_A K = K[t]/FK[t]$, and a product of fields, finite separable extensions of K ; this product (as whenever one localizes an artinian ring, here B'_K , with respect to a multiplicatively closed set) is a direct factor of B'_K , thus corresponding to a decomposition $F = F_1 F_2$ in $K[t]$, with the generator of L corresponding to t already annihilated by F_1 . Now, *since A is normal, the F_i belong to $A[t]$* (assuming them monic). Observing that $B \rightarrow L = B \otimes_A K$ is injective (since $A \rightarrow K$ is so and B is flat over A), it follows that one already has $F_1(u) = 0$, with u the class of t in L . Assuming that F was chosen of minimal degree, it follows that $F_2 = 1$ (N.B. One has $F'(u) = F'_1(u)F_2(u) + F_1(u)F'_2(u) = F'_1(u)F_2(u)$ since $F_1(u) = 0$, whence $F'_1(u) \neq 0$ since $F'(u) \neq 0$.)

Thus one has

$$(*) \quad L = B \otimes_A K = K[t]/FK[t]$$

and F is consequently a separable polynomial in $K[t]$ (but of course not necessarily in $A[t]$). (N.B. For the moment, we have essentially shown only that in 7.6 one can choose F and $\mathfrak{n}$ in such a way that—with the notation used here— $B' \rightarrow B'_{\mathfrak{n}} = B$ is *injective*; we used the normality of A for this; I do not know whether it remains true without the normality assumption.)

Let us now recall the well-known lemma taken from Serre's course of last year:

LEMMA 9.6. *Let K be a ring, $F \in K[t]$ a monic separable polynomial, $L = K[t]/FK[t]$, and u the class of t in L (so that $F'(u)$ is an invertible element of L). Then one has the formulas (where $n = \deg F$):*

$$\begin{aligned} \operatorname{tr}_{L/K} u^i / F'(u) &= 0 \quad \text{if } 0 \leq i < n-1, \\ \operatorname{tr}_{L/K} u^{n-1} / F'(u) &= 1. \end{aligned}$$

COROLLARY 9.7. *The determinant of the matrix $(u^j \cdot u^i / F'(u))_{0 \leq i, j \leq n-1}$ is equal to $(-1)^{n(n-1)/2}$, hence is invertible in every subring A of K .*

COROLLARY 9.8. *Let A be a subring of K , V the A -module generated by the u^i ($0 \leq i \leq n-1$) in L , and V' the A -submodule of L consisting of the $x \in L$ such that $\operatorname{tr}_{L/K}(xy) \in A$ for every $y \in V$ (i.e. for y of the form u^i , $0 \leq i \leq n-1$). Then V' is the A -module with basis the $u^i / F'(u)$ ($0 \leq i \leq n-1$).*

⁷Cf. EGA IV 5.8.6.

COROLLARY 9.9. *Suppose that K is the field of fractions of a normal integral domain A , with F having its coefficients in A . Then, with the notation of 9.8, V' contains the integral closure A' of A in L , which is therefore contained in $A[u]/F'(u)$ and a fortiori in $A[u][F'(u)^{-1}]$.*

Let us apply this last corollary to the situation obtained in the proof: since $F'(u)$ is invertible in B , which contains $A[u]$, B contains A' . By the Main Theorem (or from the fact that $B = A[u]_{\mathfrak{n}}$), B is a localization of A' . Since A' is normal, so is B .

Proof of (ii). One proceeds as in the preceding proof to show that, in 7.6, one can choose F so that $(*)$ still holds. The only obstacle a priori is that, since B is no longer assumed flat over A , one can no longer assert that $B \rightarrow L$ is injective, so that the argument applies a priori only to the image B_1 of B under the said homomorphism. It follows at once that B_1 is flat over A (being a localization of a free algebra over A). By 4.8, the morphism $B \rightarrow B_1$ is étale, hence an isomorphism, which completes the proof.

(From the point of view of exposition, one should interchange the last two proofs and put the formal calculations of the lemma and its corollaries in a separate no.)

COROLLARY 9.10. *Let $f: X \rightarrow Y$ be an étale morphism. If Y is normal, so is X ; the converse holds if f is surjective.*

COROLLARY 9.11. *Let $f: X \rightarrow Y$ be a dominant morphism, with Y normal and X connected. If f is net, f is étale, hence X is normal and consequently (being connected) irreducible.*

Let U be the set of points where f is étale; it is open, and it suffices to show that it is also closed and nonempty. U contains the inverse image of the generic point of Y (since, for an algebra over a field, unramified = étale), hence $(X \text{ dominating } Y)$ is nonempty. If x belongs to the closure of U , then it belongs to the closure of an irreducible component U_i of U , hence to an irreducible component $X_i \stackrel{v}{=} \bar{U}_i$ of X which meets U and consequently dominates Y (since every component of U , flat over Y , dominates Y). Thus, if y is the projection of x on Y , $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is injective (taking into account that $\mathcal{O}_y$ is an integral domain). Since $\mathcal{O}_y$ is normal and $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is net, one concludes using 9.5(ii).

COROLLARY 9.12. *Let $f: X \rightarrow Y$ be a dominant morphism of finite type, with Y normal and X irreducible. Then the set of points where f is étale is identical to the complement of the support of $\Omega_{X/Y}^1$, i.e. to the complement of the subscheme of X defined by the different ideal $\mathfrak{d}_{X/Y}$.*

(This is the “less trivial” statement alluded to in the remark in no. 4.)

REMARK. One should beware of believing that a connected étale covering of an irreducible scheme is itself irreducible when the base is not assumed normal. This question will be studied in no. 11.

10. Étale coverings of a normal scheme

PROPOSITION 10.1. *Let X be an étale separated prescheme over a normal connected Y with field K . Then the connected components X_i of X are integral, their fields K_i are finite separable extensions of K , and X_i is identified with a nonempty open subset of the normalization of X in K_i (hence X with a dense open subset of the normalization of Y in $R(X) = L = \prod K_i$).*

By 9.10, X is normal; a fortiori its local rings are integral domains, so the connected components of X are irreducible. Since X_i is normal, and finite and dominant over Y , it follows from a special case (moreover, almost trivial) of the Main Theorem that X_i is an open subset of the normalization of X in the field K_i of X .

COROLLARY 10.2. *Under the conditions of 10.1, X is finite over Y (i.e. an étale covering of Y) if and only if X is isomorphic to the normalization Y' of Y in $L = R(X)$ (the ring of rational functions on X).*

Indeed, one knows that this normalization is finite over Y (Y being normal and R/K separable); conversely, if X is finite over Y , it is finite over Y' , so its image in Y' is closed, and on the other hand it is dense.

An algebra L of finite rank over K will be called *unramified over X* (or simply unramified over K , if X is understood) if L is a separable algebra over K , i.e. a direct product of separable extensions K_i , and if the normalization Y' of Y in L (the disjoint sum of the normalizations of Y in the K_i) is unramified (= étale by 9.11) over Y . Thus:

COROLLARY 10.3. *For every X finite over Y whose every irreducible component dominates Y , let $R(X)$ be the ring of rational functions on X (the product of the local rings of the generic points of the irreducible components of X), so that $X \mapsto R(X)$ is a functor with values in the algebras of finite rank over $K = R(Y)$. This functor establishes an equivalence of the category of connected étale coverings of Y with the category of extensions L of K unramified over Y .*

The inverse functor is the normalization functor.

Suppose that Y is affine, hence defined by a normal ring A with field of fractions K . Let L be a finite extension of K which is a direct product of fields; then, by definition, the normalization Y' of Y in L is isomorphic to $\text{Spec}(A')$, where A' is the normalization of A in L . To say that L is unramified over Y means that A' is unramified (or equivalently, étale) over A . If A is local, this amounts to saying that the local rings $A'_\mathfrak{n}$ (where $\mathfrak{n}$ runs through the finite set of maximal ideals of A' , i.e. its prime ideals inducing the maximal ideal $\mathfrak{m}$ of A) are unramified (= étale) over the local ring A . Finally, note also that the discriminant criterion (4.10) can also be applied in this situation (more generally, a variant of the said criterion should be stated as follows, without a preliminary flatness condition when X dominates Y , with Y nevertheless assumed locally integral: $A \rightarrow B$ and $B \rightarrow B \otimes_A K = L$ are injective—then $\text{tr}_{L/K}$ is defined—and $\text{tr}_{L/K}(xy)$ induces a *fundamental bilinear form* $B \times B \rightarrow A$, i.e. there exist $x_i \in B$ ($1 \leq i \leq n$, $n = \text{rank of } L \text{ over } K$) such that $\text{tr}(x_i x_j) \in A$ for every i, j and $\det(\text{tr}(x_i x_j))$ is invertible in A).

The sorites (4.6) immediately implies the sorites of unramifiedness in the classical setting:

PROPOSITION 10.4. *Let Y be a normal integral prescheme, with field K . (i) K is unramified over Y . (ii) If L is an extension of K unramified over Y , if Y' is a normal prescheme with field L dominating Y (for example, the normalization of Y in L), and M an extension of L unramified over Y' , then M/K is unramified over X (transitivity of unramifiedness). (iii) Let Y' be a normal integral prescheme dominating Y , with field K'/K ; if L is an extension of K unramified over Y , then $L \otimes_K K'$ is an extension of K' unramified over Y' (the translation property).*

Moreover:

COROLLARY 10.5. *Under the conditions of (iii), if $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, then the normalization $\bar{A}'$ of Y' in $L' = L \otimes_K K'$ is identified with $\bar{A} \otimes_A A'$, where $\bar{A}$ is the normalization of A in L .*

Usually, people (who are reluctant to consider rings which are not integral domains, even if they are direct products of fields) state the translation property in the following (weaker) form:

COROLLARY 10.6. *Under the conditions of (iii), let L_1 be a compositum of L/K (unramified over Y) and K'/K . Then L_1/K' is unramified over Y' . When $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, one moreover has*

$$\bar{A}' = A[\bar{A}, A']$$

i.e. the ring $\overline{A'}$, the normalization of A' in L_1 , is the A -algebra generated by A' and the normalization $\overline{A}$ of A in L .

This last fact is false without the unramifiedness assumption, even in the case of composita of number fields...

To conclude this no., we shall give the interpretation of the notion of étale covering corresponding to its intuitive picture: there should be the “maximum number” of points above the point $y \in Y$ under consideration, and in particular there should not be “several coincident points” above y . To prove results in this direction with all the desired generality, we shall admit Proposition 10.7 below (whose proof will appear in the multiplodoque, Chap. IV, par. 15, and uses Chevalley’s technique of constructible sets, and a little descent theory...).

A morphism of finite type $f: X \rightarrow Y$ is said to be *universally open* if for every extension of the base $Y' \rightarrow Y$ (with Y' locally noetherian) the morphism $f': X' = X \times_Y Y' \rightarrow Y'$ is open, i.e. sends open sets to open sets. One can, moreover, restrict oneself to the case where Y' is of finite type over Y (and even where Y' is of the form $Y[t_1, \dots, t_r]$, with the t_i indeterminates). A universally open morphism is a fortiori open (the converse being false); on the other hand, if f is open, with X and Y irreducible, then all the components of all the fibers of f have the same dimension (namely, the dimension of the generic fiber $f^{-1}(z)$, with z the generic point of Y). Finally, if Y is normal, this last condition already implies that f is *universally open* (Chevalley’s theorem). It follows, for example, that if $f: X \rightarrow Y$ is a *quasi-finite* morphism, with Y normal and irreducible, then f is universally open (or equivalently, open) if and only if every irreducible component of X dominates Y . Recall also that a flat morphism (of finite type), being open, is also universally open. With these preliminaries in place, let us “recall” the following

PROPOSITION 10.7. *Let $f: X \rightarrow Y$ be a quasi-finite separated universally open morphism. For every $y \in Y$, let $n(y)$ be the “geometric number of points of the fiber $f^{-1}(y)$ ”, equal to the sum of the separable degrees of the residue field extensions $k(x)/k(y)$, for the points $x \in f^{-1}(y)$. Then the function $y \mapsto n(y)$ on Y is upper semicontinuous. For it to be constant in a neighborhood of the point y (i.e. for one to have $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y containing y), it is necessary that there exist a neighborhood U of y such that $X|U$ is finite over U .⁸*

COROLLARY 10.8. *If $y \mapsto n(y)$ is constant and Y is geometrically unibranch⁹, the irreducible components of X are disjoint.*

PROPOSITION 10.9. *Let $f: X \rightarrow Y$ be an étale separated morphism. With the notation of 10.7, the function $y \mapsto n(y)$ is upper semicontinuous. For it to be constant in a neighborhood of the point y (i.e. for one to have $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y containing y), it is necessary and sufficient that there exist an open neighborhood U of y such that $X|U$ is finite over U , i.e. is an étale covering of U .*

COROLLARY 10.10. *For a separated étale morphism $f: X \rightarrow Y$, with Y connected, to be finite (i.e. to make X an étale covering of Y), it is necessary and sufficient that all the fibers of f have the same geometric number of points.*

In 10.7 and its corollary, there was no normality assumption on Y . If one makes such an assumption, one obtains the stronger statement (most often taken as the definition of a covering being unramified):

THEOREM 10.11. *Let $f: X \rightarrow Y$ be a quasi-finite separated morphism. Suppose that Y is irreducible, that every component of X dominates Y , and that X is reduced (i.e. $\mathcal{O}_X$ has no nilpotent elements). Let n be the degree of X over Y (the sum of the degrees, over the field K of Y , of the fields K_i of the irreducible components X_i of X). Let y be a normal point*

⁸Cf. EGA IV 15.5.1.

⁹For the definition, cf. below, no. 11, p. 15.

of Y . Then the geometric number $n(y)$ of points of X above y is $\leq n$, with equality if and only if there exists an open neighborhood U of y such that $X|U$ is an étale covering of U .

The “only if” being trivial, let us prove the “if”. Let z be the generic point of Y ; one has $n(z) = (\text{sum of the separable degrees of the } K_i/K) \leq n$, and by 10.7, $n(y) \leq n(z) \leq n$, with equality implying that $X|U$ is *finite* over U for a suitable neighborhood U of y . One may therefore suppose that X is finite over Y and that the function $n(y')$ on Y is constant. Finally, by 10.8, X is then the disjoint union of its irreducible components, and to prove that it is unramified at y , one is reduced to the case where X is irreducible, hence integral. Finally, one may suppose that $Y = \text{Spec}(\mathcal{O}_y)$. The theorem then reduces to the following classical statement:

COROLLARY 10.12. *Let A be a normal local ring (noetherian, as always) with field K , L a finite extension of K of degree n and separable degree n_s , B a subring of L finite over A , with field of fractions L , $\mathfrak{m}$ the maximal ideal of A , and n' the separable degree of $B/\mathfrak{m}B$ over $A/\mathfrak{m}A = k$ (= the sum of the separable degrees of the residue field extensions of this ring). One has $n' \leq n_s$ and a fortiori $n' \leq n$. This last inequality is an equality if and only if B is unramified (= étale) over A .*

It remains only to show that $n' = n$ implies that B is étale over A . Let us recall the proof when k is infinite: one need only show that $R = B/\mathfrak{m}B$ is separable over k ; if this were not so, it would follow (by a known lemma) that there exists an element a of R whose minimal polynomial over k has degree $> n'$. This element comes from an element x of B , whose minimal polynomial over K (as an element of L) has degree $\leq n$; on the other hand, the latter has its coefficients in A since A is normal, and hence gives, by reduction mod $\mathfrak{m}$, a monic polynomial $F \in k[t]$ of degree $\leq n = n'$ such that $F(a) = 0$, a contradiction.

In the general case (k possibly finite), returning to geometric language, one considers $Y' = \text{Spec}(A[t])$, which is faithfully flat over Y , and the generic point y' of the fiber $\text{Spec}(k[t])$ of Y' over y . Then X is net over Y at y if and only if $X' = X \times_Y Y' = \text{Spec}(B[t])$ is net at y' over Y' , as one sees at once. Moreover, by the choice of y' , its residue field is $k(t)$, hence infinite. Since y' is a normal point of Y' , one is reduced to the preceding case.

11. Some complements

We have already said that a connected étale covering of an integral scheme is not necessarily integral. Here are two examples of this fact.

a) Let C be an algebraic curve with an ordinary double point x , C' its normalization, and a and b the two points of C' above x . Let C'_i ($i = 1, 2$) be two copies of C' , and a_i and b_i the point of C'_i corresponding to a resp. b . In the sum curve $C'_1 \amalg C'_2$, identify a_1 with b_2 , and a_2 with b_1 (we leave it to the reader to make the identification process precise; it will be explained in Chap. VI of the multiplodoque but, in the case of curves over an algebraically closed field, is treated in Serre’s book on algebraic curves). One obtains a *connected* and *reducible* curve C'' which is an étale covering of degree 2 of C . The reader will verify that, in general, the connected “Galois” étale coverings C'' of C whose inverse image $C'' \times_C C'$ is a *trivial* covering of C' (i.e. isomorphic to the sum of a certain number of copies of C') are “cyclic” of degree n , and that for every integer $n > 0$ one can construct a connected cyclic étale covering of degree n . In the language of the fundamental group, which will be developed later, this means that the quotient of $\pi_1(C)$ by the closed normal subgroup generated by the image of $\pi_1(C') \rightarrow \pi_1(C)$ (the homomorphism induced by the projection) is isomorphic to the profinite completion of $\mathbf{Z}$. More precisely, one should be able to show that the fundamental group of C is isomorphic to the (topological) free product of the fundamental group of C' with the profinite completion of $\mathbf{Z}$. Note that questions of this kind gave rise to “descent theory” for schemes.

b) Let A be a complete local integral domain. One knows that its normalization A' is finite over A (Nagata), hence is a complete semilocal ring, and therefore local since it is an

integral domain. Suppose that the residue field extension L/k which it defines is not purely inseparable (otherwise, one will say that A is geometrically unibranch; cf. below). This is the case, for example, for the ring $\mathbf{R}[[s, t]]/(s^2 + t^2)\mathbf{R}[[s, t]]$, where $\mathbf{R}$ is the field of real numbers. Let k' be a finite Galois extension of k such that $L \otimes_k k'$ decomposes, and let B be a finite étale algebra over A corresponding to the residue field extension k' (recall that B is essentially unique). Then $B' = A' \otimes_A B$ over B has residue algebra $L \otimes_k k'$, which is not local; hence B' is not a local ring, and therefore (being complete) has zero divisors. Since it is contained in the total ring of fractions of B (being free over A' , hence torsion-free over A' , hence torsion-free over A , and therefore contained in $B' \otimes_A K = B'_{(K)} = A'_{(K)} \otimes_K B_{(K)} = B_{(K)}$ since $A'_{(K)} = K$), it follows that B is not an integral domain. In the case of the ring $\mathbf{R}[s, t]/(s^2 + t^2)\mathbf{R}[s, t]$, taking $k'/k = \mathbf{C}/\mathbf{R}$, one obtains for B the local ring of two intersecting lines in the plane at their point of intersection.

Note also that if there exists a connected étale covering X of an integral Y which is not irreducible, then every irreducible component of X gives an example of an unramified covering X' of Y , dominating Y , which is not étale over Y . In example a), one thus finds that C' is unramified over C , without being étale at the two points a and b (as one also sees directly by inspecting the completions of the local rings of x and a : from the “formal” point of view, C' at the point a is identified with a closed subscheme of C at the point x , namely one of the two “branches” of C through x).

In a) and b), one sees that the failure of the conclusions of 9.5. (i) and (ii) is directly linked to the fact that a point of Y “splits” into *distinct* points of the normalization (in b), the fact that the residue field extension is not purely inseparable must be interpreted geometrically in this way). Precisely, we shall say that a local integral domain A is *geometrically unibranch* if its normalization has only one maximal ideal, with the corresponding residue field extension purely inseparable; a point y of an integral prescheme is said to be geometrically unibranch if its local ring is so. Examples: a normal point, an ordinary cusp of a curve, etc. It seems that if Y has a point which is not unibranch, there always exists a connected étale covering of Y which is not irreducible; at least this is what we have shown in case b), when Y is the spectrum of a complete local ring. One can show, on the other hand, that *if all points of Y are geometrically unibranch, then every connected unramified Y -prescheme dominating Y is étale and irreducible*. The proof follows that of 9.5, using the following generalization of Theorem 8.3, which will be proved later by means of descent techniques¹⁰:

Let $Y' \rightarrow Y$ be a finite, radicial, surjective morphism (i.e. what one might call a “universal homeomorphism”). Consider the functor $X \mapsto X \times_Y Y' = X'$ from Y -preschemes to Y' -preschemes. This functor induces an equivalence of the category of étale Y -schemes with the category of étale Y' -schemes. One can apply this result, for example, when Y' is the normalization of Y , with Y assumed unibranch (and Y' finite over Y , which holds in all cases encountered in practice), or to a Y'' “sandwiched” between Y and its normalization (which no longer needs to be finite over Y).

¹⁰Cf. IX 4.10. For a more direct proof, cf. EGA IV 18.10.3, using a variant of 9.5 for geometrically unibranch local rings.